Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

[Project Report]

ATM Simulator

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Project Overview:

The ATM Simulator is a console-based Java application designed to replicate the basic operations of an Automated Teller Machine. The main aim of this project is to provide a simple and user-friendly system where users can securely access their account using a Personal Identification Number (PIN) and perform essential banking operations such as balance inquiry, deposit, and withdrawal.

This project addresses the need for understanding real-world banking transaction logic using core Java programming concepts. The system demonstrates the practical implementation of object-oriented programming, control structures, and input handling in Java. The project is significant for beginners as it bridges theoretical programming knowledge with a real-life inspired application.

Objective:

The specific objectives of this project are:

- To design and implement a console-based ATM simulation system
- To apply object-oriented programming concepts in Java
- To implement secure PIN-based user authentication
- To develop menu-driven banking operations
- To perform deposit and withdrawal with proper validation
- To handle user input efficiently using the Scanner class
- To strengthen problem-solving and logical thinking skills
- To gain hands-on experience in Java application development

Literature Review:

Automated Teller Machines (ATMs) are widely used electronic banking systems that allow customers to perform financial transactions without direct interaction with bank staff. Modern ATM systems provide services such as cash withdrawal, balance inquiry, fund transfer, and mini statements.

Previous academic and tutorial-based ATM simulation projects mainly focus on implementing authentication mechanisms, transaction processing, and menu-driven interfaces using programming languages such as Java and C++. According to Java programming tutorials from GeeksforGeeks and Oracle documentation, console-based simulations are effective for teaching core programming and object-oriented concepts.
This project follows standard Java programming practices to build a simplified ATM model suitable for educational purposes while maintaining logical accuracy of real ATM operations.

Methodology:

The development of the ATM Simulator will follow a structured approach:

1. System Design
   - Identify functional requirements
   - Design class structure (ATM/BankAccount)
   - Plan user interaction flow

2. Testing
   - Test correct and incorrect PIN scenarios
   - Test deposit and withdrawal operations
   - Test boundary conditions
   - Debug and refine the program

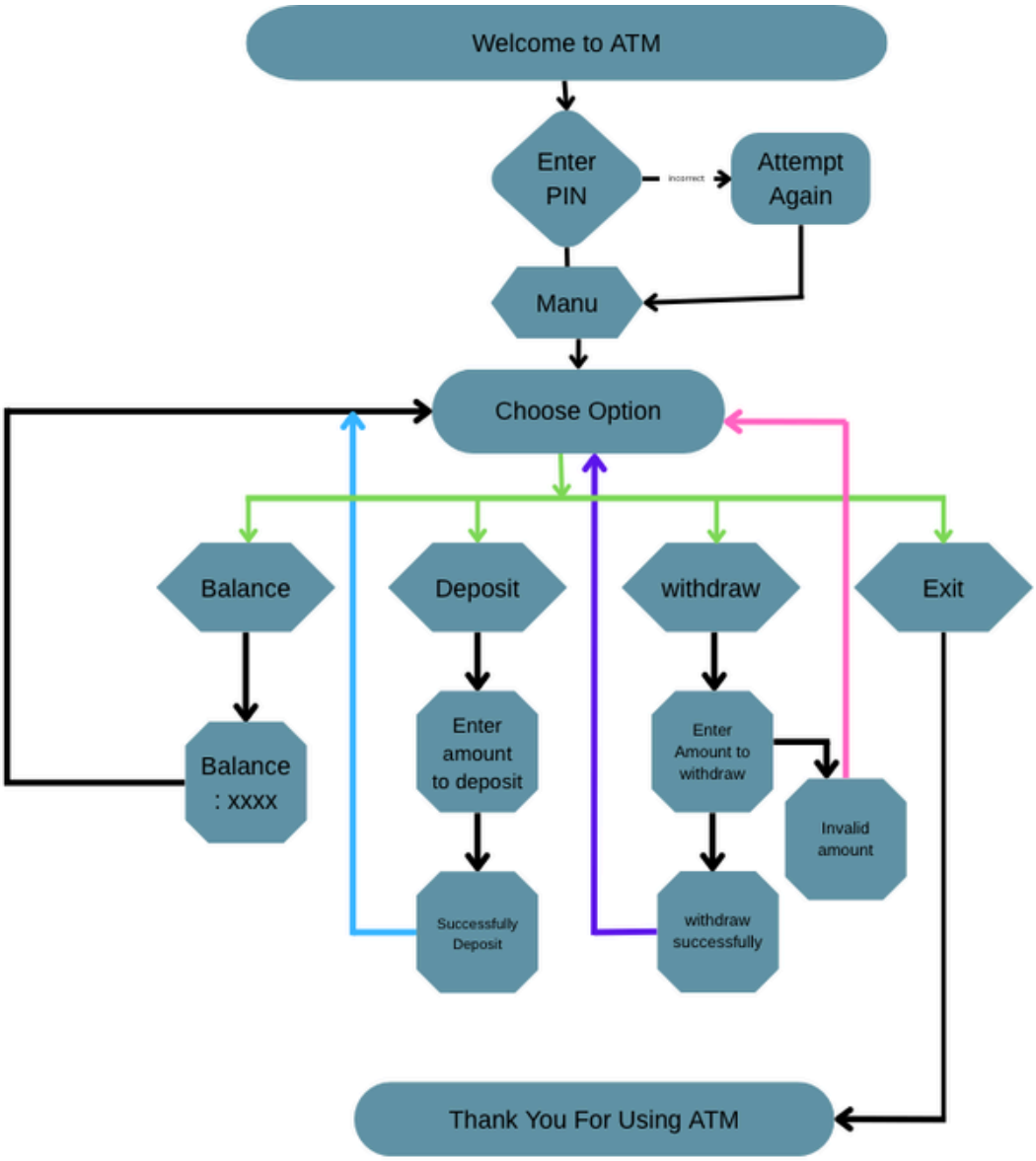

Fig-1 : ATM Simulator Flowchart

Implementation

1. Design
   - Basic design
   - Enter pin then choose options
   - Get information or deposit or withdraw money

```java
int pin = 1223;
double balance = 5000;
int attempts = 3;
```

Fig 2 :: Set pin and attempt

```java
System.out.println(x: "Welcome to ATM");

while (attempts > 0) {
    System.out.print(s: "Enter PIN: ");
    int enteredPin = sc.nextInt();
```

Fig 3: Enter pin

```java
} else {
    attempts--;
    System.out.println("Wrong PIN. Attempts left: " + attempts);
}
```

Fig 4: wrong attempts

```java
if (attempts == 0) {
    System.out.println(x: "Account locked");
}
```

Fig 5: Account locked (after 3 attempts)

```java
if (enteredPin == pin) {
    int choice;

    do {
        System.out.println(x: "\n1. Check Balance");
        System.out.println(x: "2. Deposit Money");
        System.out.println(x: "3. Withdraw Money");
        System.out.println(x: "4. Exit");
        System.out.print(s: "Choose option: ");
        choice = sc.nextInt();
```

Fig 6: Entering correct pin go to manu

Result and Analysis:

- Upon successful running we go to Welcome to ATM
- Then we have to enter our pin
- if we enter wrong pin we get a Attempts left :2 massage
- After that if we enter wrong 2 more times the account will be locked
- But if we enter the correct pin we go to manu
- In manu choosing option 1 checks Balance
- Choosing option 2 we can deposit money to the account
- Choosing option 3 we can withdraw money but if we withdraw more then what we have we get a Invalid account massage and go back to manu
- Choosing option 4 we can exit from the ATM

```
Welcome to ATM
Enter PIN:
```

Fig 7: Enter PIN

```
Welcome to ATM
Enter PIN: 4567
Wrong PIN. Attempts left: 2
Enter PIN:
```

Fig 8: Entering wrong pin

```
Welcome to ATM
Enter PIN: 4567
Wrong PIN. Attempts left: 2
Enter PIN: 678
Wrong PIN. Attempts left: 1
Enter PIN: 789
Wrong PIN. Attempts left: 0
Account locked
PS E:\project>
```

Fig 9: Account locked(3 wrong attemps)

```
Welcome to ATM
Enter PIN: 1223

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Choose option:
```

Fig 10: Entering Right pin (Enter manu)

```
Choose option: 1
Balance: 5000.0

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Choose option:
```

Fig 11: Balance Check

```
Choose option: 2
Enter amount to deposit: 600
Deposited successfully

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Choose option:
```

Fig 12: Deposit Money

```
Choose option: 3
Enter amount to withdraw: 500
Withdraw successful

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Choose option:
```

Fig 13: Withdraw Money

```
Choose option: 3
Enter amount to withdraw: 6000
Insufficient balance or invalid amount

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
```

Fig 14: invalid amount (if balance is low)

```
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Choose option: 4
Thank you for using ATM
PS E:\project>
```

Fig 15: Exit

Discussion:

1. Challenges
   - I was facing some issues with my Vs code editor not running
   - I was also facing issues with github because i have never used it before so i had to learn it fully
   - with each challenged i faced i learned new things

Conclusion:

This project was all about how i created a simpler version of ATM Simulator using java . this can be a very useful thing in real life, because it has a limited pin trail which if 3 wrong pin were attempted the system will be locked . Its a very usefull feature if someone is trying to get into your account.
The design isn't perfect , there are a lot of things that can be improved . Like the interface can be a bit better and we can also add alart feature , when someone attempts 3 wrong pin the account owner gets a notification

References:

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